

FreeCRM Automation Testing Framework

Selenium WebDriver • Java • Cucumber BDD • Maven

Project Documentation

Automation Testing Framework for FreeCRM CRM Application.

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1. Introduction

Sprint 0 represents the preparation phase of the automation project where the team sets up the testing environment, designs the framework architecture and prepares the backlog before starting automation development.

This automation framework is built for the FreeCRM web application using Selenium WebDriver, Java and Cucumber BDD following the Page Object Model design pattern.

2. Objectives of Sprint 0

- Understand FreeCRM application workflow
- Setup automation testing environment
- Design framework architecture
- Create feature files and step definitions
- Prepare product backlog
- Assign responsibilities among team members

3. Project Overview

FreeCRM is a Customer Relationship Management system used to manage contacts, deals, companies, tasks and cases. The automation framework validates these modules using automated UI tests.

Main Functionalities Tested

- Login functionality
- Contacts management
- Deals module
- Companies module
- Cases module
- Tasks module
- Logout functionality

4. Tools and Technologies

Tool	Purpose
Selenium WebDriver	Automates browser actions
Java	Programming language used for automation scripts
Cucumber	BDD framework for writing scenarios

Maven	Project dependency management and build
Eclipse IDE	Development environment
Chrome Browser	Used for test execution
Page Object Model	Improves maintainability of test code

5. Automation Framework Architecture

Feature Files



Step Definitions



Page Classes (POM)



Utilities / DriverFactory



Selenium WebDriver



FreeCRM Web Application

This layered architecture improves maintainability and separates test logic from UI elements.

6. Project Structure

FreeCRM_Automation_Framework

■■■ src/main/java/pages

■■■ src/main/java/util

■■■ src/test/java/stepdefinitions

■■■ src/test/java/tests

■■■ src/test/resources/features

■■■ src/test/resources/Testdata

■■■ Testreports

■■■ pom.xml

7. Product Backlog

Story ID	User Story	Priority
US01	User should be able to login	High
US02	User manages contacts	High
US03	User manages deals	High
US04	User manages companies	Medium

US05	User manages tasks	Medium
US06	User logs out	Medium

8. Scrum Board (Sprint 0)

To Do	In Progress	Done
Project Setup		Environment Setup
Framework Design		Maven Configuration
Identify Test Scenarios		POM Setup
Create Feature Files		Step Definitions
Configure WebDriver		Config Setup

9. BDD Feature Example

Feature: FreeCRM Login

Scenario: Login with valid credentials

Given user is on login page

When user enters valid username and password

Then user should be logged into the dashboard

10. Sprint 1 – Sample Test Cases

Test Case ID	Scenario	Expected Result
TC01	Login with valid credentials	Dashboard displayed
TC02	Open contacts module	Contacts list visible
TC03	Create new deal	Deal saved successfully
TC04	Open companies module	Companies page visible
TC05	Logout	User redirected to login page

11. Automation Test Plan

Component	Description
Scope	Automation of login, contacts, deals, companies and tasks
Framework	Selenium + Java + Cucumber BDD
Execution	Tests executed through TestRunner
Reports	HTML test reports generated after execution